Section 1 - Question 2 - COE Price Prediction Model

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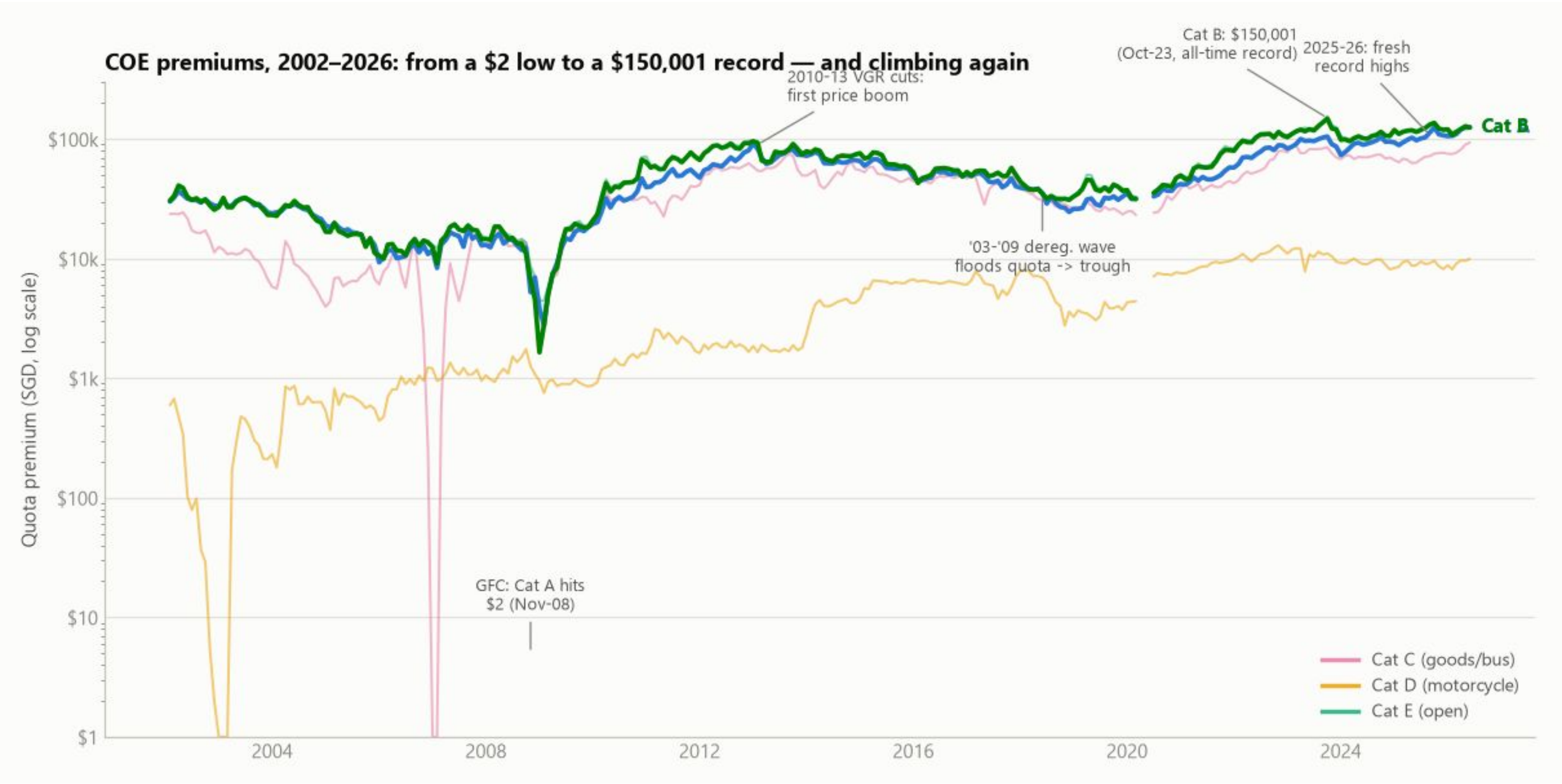
Objectives

1. **Evaluate whether future Category A and B COE prices can be accurately predicted** using only information available before each bidding exercise.

2. **Quantify the independent impact of COE quota on prices**, after controlling for other market and economic factors.

3. **Estimate the price sensitivity to quota changes**, providing actionable estimates (e.g. the impact of an additional 100 COEs) for policy planning.

Exploratory



Key Milestones in Singapore's COE Market

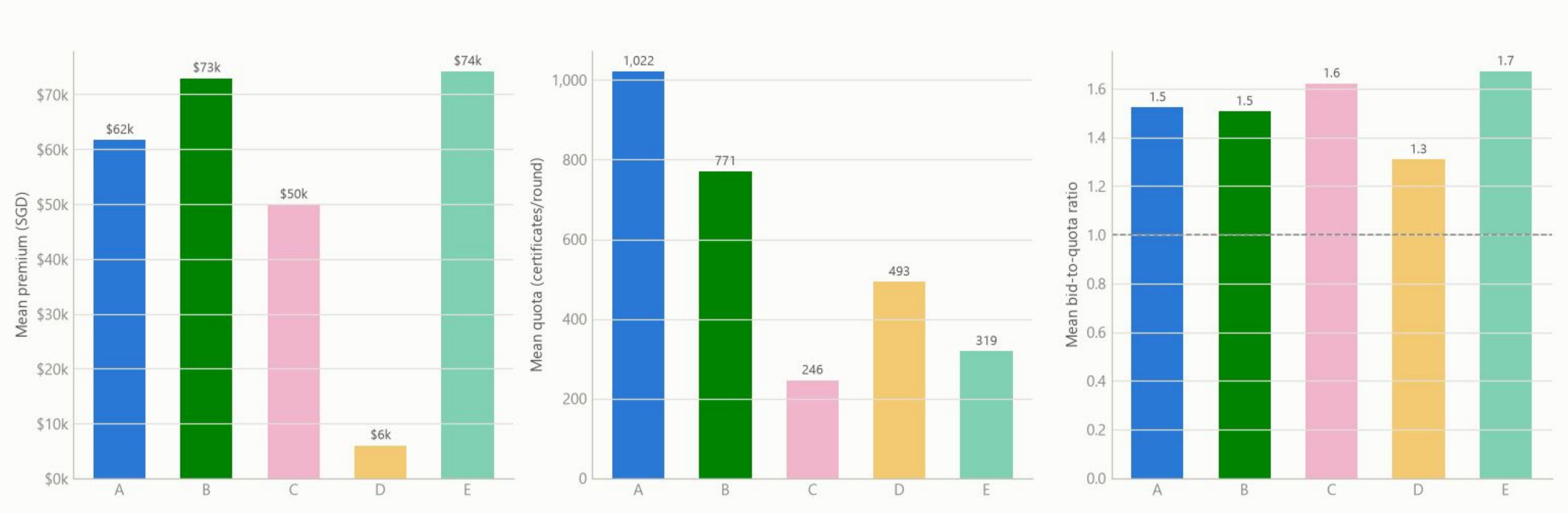
1. 2008–2009 (Global Financial Crisis): Demand collapsed while quota increased from expiring COEs, driving Category A prices to a low of S$2.

2. 2010–2013: Progressive reductions in the Vehicle Growth Rate (3% → 0.5%) tightened COE supply, resulting in a sharp rise in prices.

3. 2014–2019: A surge in COE expiries expanded quota, leading to a sustained six-year decline in COE prices.

4. 2022–Present: Strong post-COVID demand combined with lower COE supply pushed prices to record highs, with Category B reaching S$150,001 (Oct 2023) and new records continuing into 2025–2026.

Exploratory



1. COE categories exhibit distinct market dynamics—price, quota size, and demand pressure do not move together.

2. Category A has the largest quota, Category C has the smallest quota despite only mid-range prices and consistently high demand pressure.

3. Category D is the only category where low prices align with low demand pressure, suggesting different demand behaviour across categories.

4. **These differences justify modelling each COE category separately, rather than applying a single model across all categories.**

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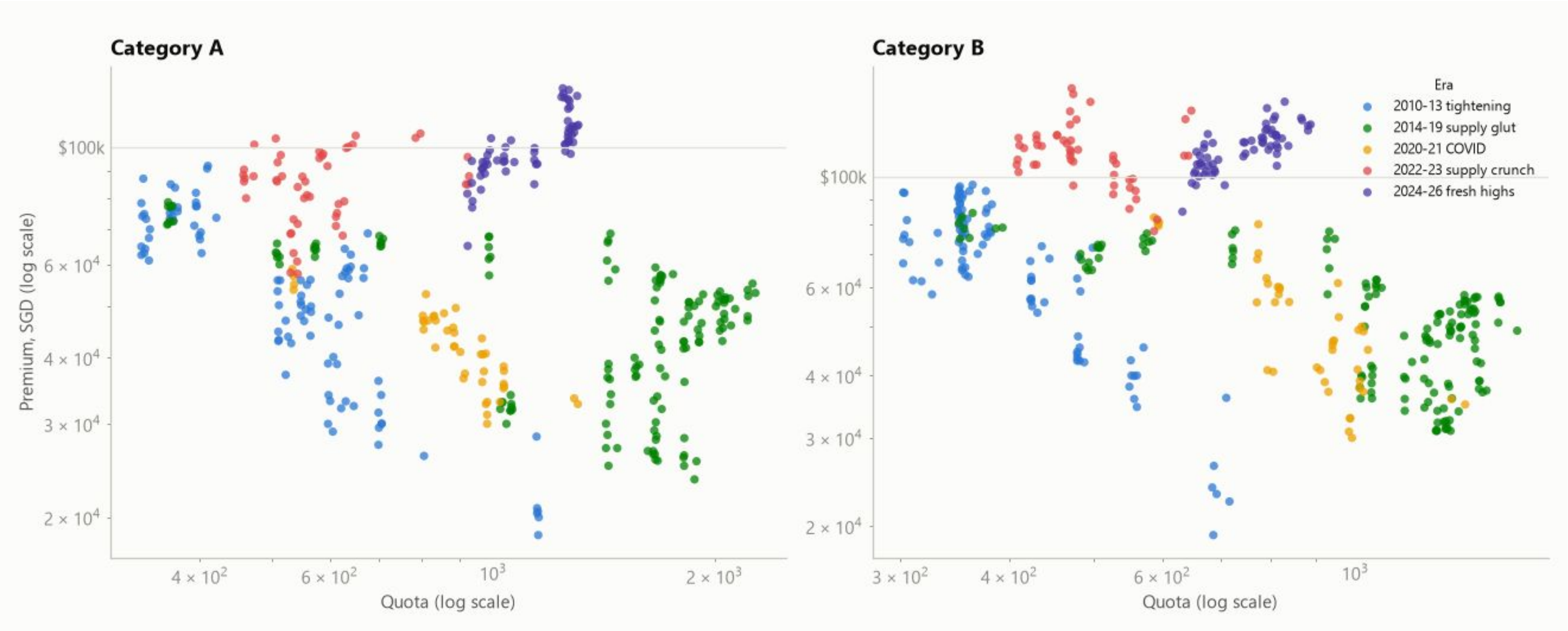
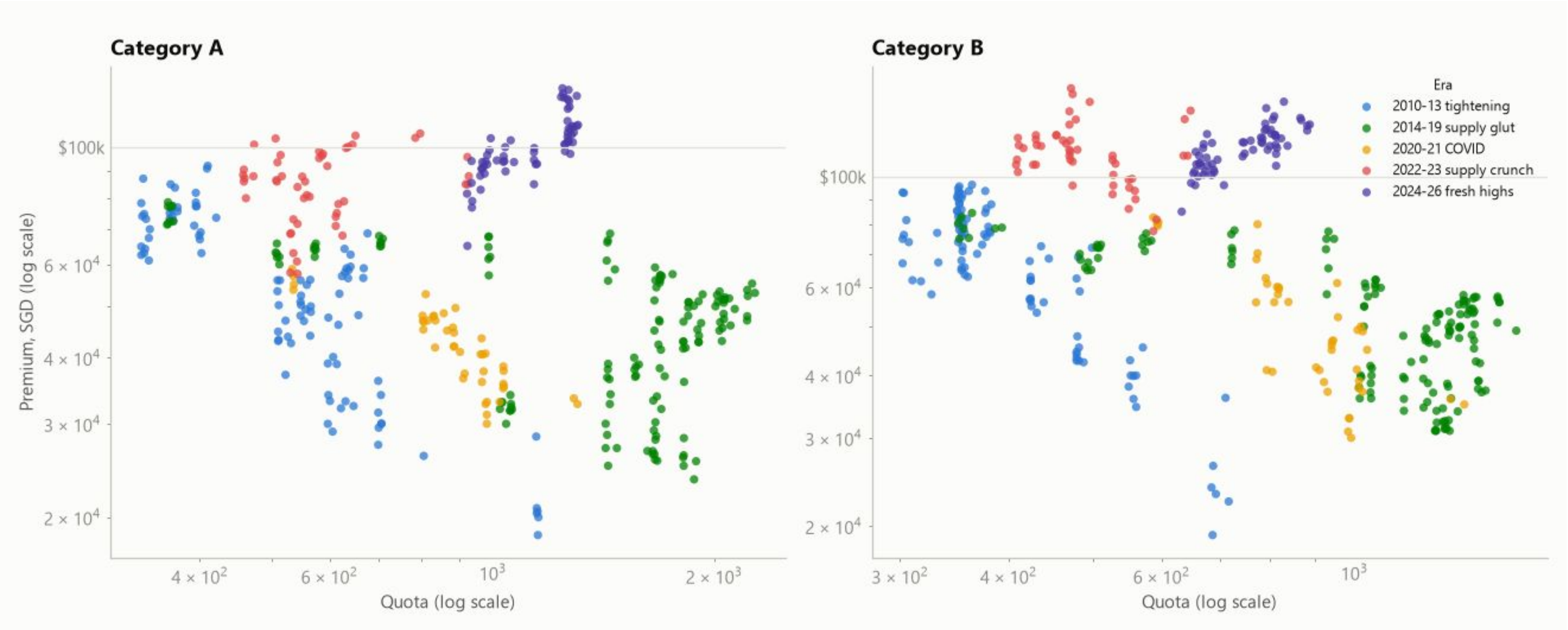
Exploratory



1. The quota–price relationship differs across policy periods, rather than following a single consistent trend.
2. Within each policy era, higher quotas are associated with lower COE prices, as expected.
3. Differences between policy eras are much larger than variations within an era, meaning historical shifts can mask the true impact of quota.
4. This motivates controlling for policy era (or year) in the regression model to isolate the true effect of quota on COE prices.

FRAMING THE QUESTION

Exploratory



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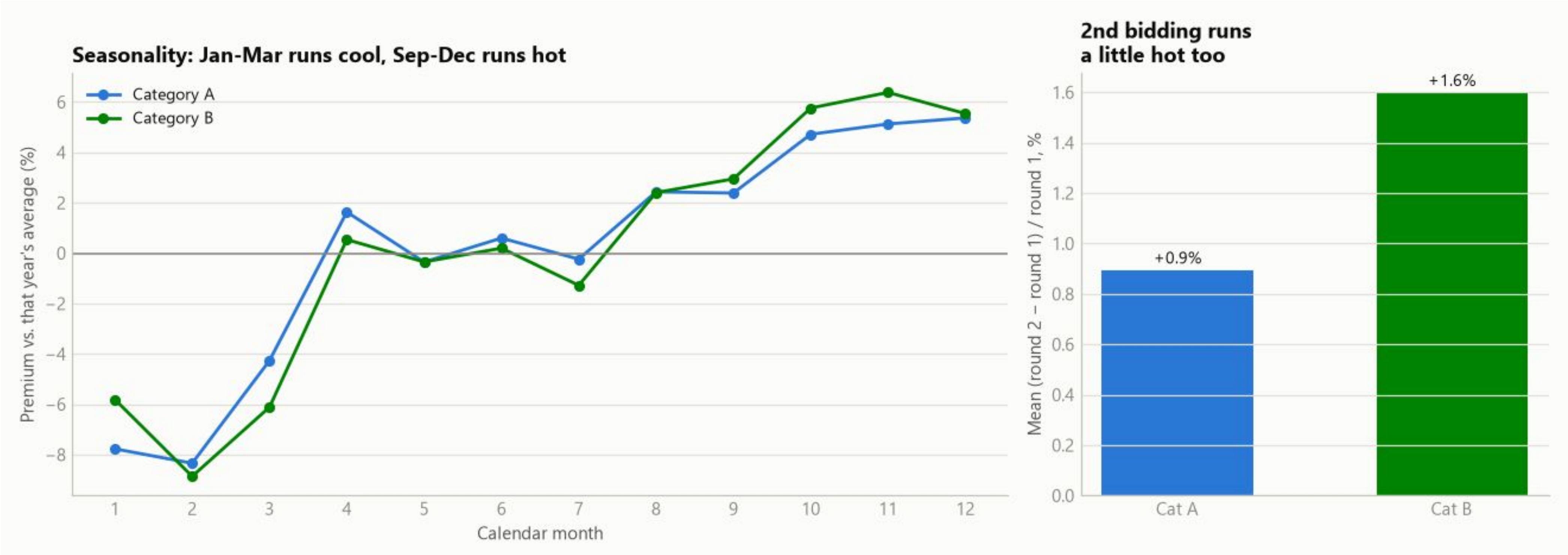


Left: COE prices exhibit consistent seasonal patterns, with prices typically 4–9% below the annual average in Jan–Mar and 2–6% above in Sep–Dec.

Right: The second bidding exercise each month consistently clears at slightly higher prices than the first (+0.9% for Category A, +1.6% for Category B).

Both seasonal and bidding-round effects are included as control variables to isolate the true impact of quota on COE prices.

Exploratory

Nominal vs. real: how much of the "record high" is just inflation?

- COE prices have increased substantially in real terms, not just because of inflation.
- Since 2010, Category A prices rose +280% nominal (+180% real) and Category B +200% nominal (+120% real), with roughly two-thirds of the increase reflecting genuine market effects.
- Adjusting for inflation has little impact on the quota elasticity estimates, as the regression model already controls for year-specific effects.

Explanatory Modelling

Estimating COE Quota Elasticity

- **Objective**: Quantify how a 1% change in COE quota affects COE prices, while holding demand and other factors constant.

- **Method**: A log-log regression is used so the coefficient directly represents the price elasticity of quota.

- **Model development**: Four progressively richer models are built by adding controls for demand (Bid-to-Quota ratio), year effects, seasonality (quarter), and bidding round.

- **Robust estimation**: Newey–West (HAC) standard errors are used to account for correlation between consecutive bidding exercises, providing more reliable statistical inference.

Explanatory Modelling

Model	Category A elasticity	Category A R^2	Category B elasticity	Category B R^2
M1 plain (ln_premium ~ ln_quota)	−0.251	0.10	−0.461	0.25
M2 + demand signal (ln_btq)	−0.235	0.10	−0.463	0.25
M3 + year control	−0.313	0.94	−0.432	0.93
M4 final (+ quarter, round)	−0.310	0.95	−0.436	0.94

1. A naive regression (M1) performs poorly, explaining only 10% (Category A) and 25% (Category B) of price variation due to multi-year market cycles.
2. Adding year controls (M3) captures these market-wide effects, increasing model fit dramatically to 93–95% R^2 for both categories.
3. The corrected model also removes bias in the elasticity estimate, with a substantial adjustment for Category A and only a minor change for Category B.

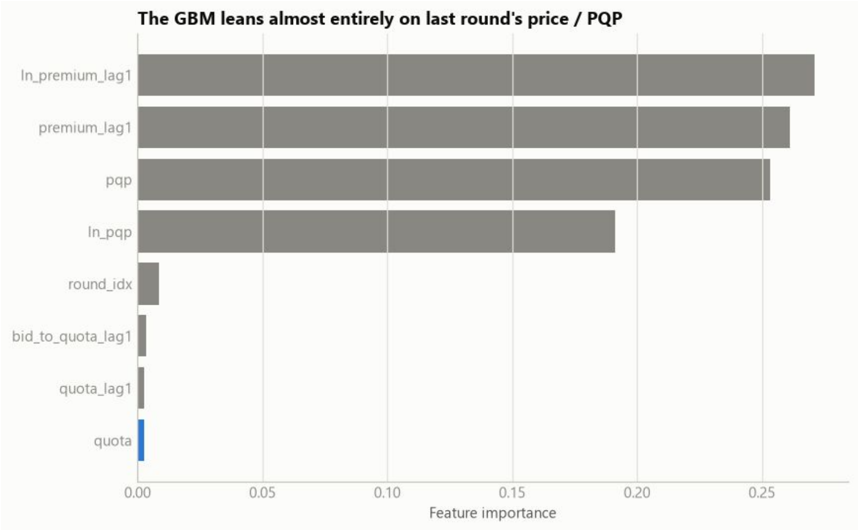
Explanatory Modelling

Category	Elasticity	95% CI	p-value	R^2	n
Category A	-0.310	[-0.40, -0.22]	<0.001	0.945	391
Category B	-0.436	[-0.61, -0.26]	<0.001	0.937	391

Robustness check	Result
Differencing (Δ instead of levels)	-0.12 (A) / -0.20 (B) -- same sign, significant
Real vs. nominal (CPI-deflated)	-0.30 vs -0.31 (A), -0.41 vs -0.44 (B) -- barely moves
Category interaction test	B is +0.11 steeper than A (p=0.003) -- real difference
Cross-category spillover	Not significant (p=0.80 / 0.18) -- treat as separate levers

Explanatory Modelling

Model	MAE	RMSE	MAPE %	R^2
1. Naive (last round)	4,036	5,294	3.61	0.800
2. PQP as predictor	3,987	5,257	3.55	0.803
3. Gradient Boosting	4,868	6,146	4.28	0.730
4. Hybrid (persistence + elasticity)	4,203	5,429	3.75	0.789



1. Simple baselines (e.g. last round's price or PQP) outperform more complex machine learning models for short-term COE price forecasting.
2. COE prices behave similarly to a random walk, making future prices inherently difficult to predict beyond recent market prices.
3. Quota has limited value for next-round prediction once recent prices are known, but it remains a significant driver of the overall COE price level.
4. Recommendation: Use PQP/last round's price for operational forecasting, and the elasticity model for evaluating the impact of quota policy changes.

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Back testing

To validate the elasticity model, we compared its predictions against a real quota increase introduced by LTA in May 2023.

Category	Quota change	Model-implied price change	Actual price change
Category A	+11.1%	-3.2%	-3.5%
Category B	+5.9%	-2.5%	-1.1%

1. **Category A**: The model closely matched reality. An 11.1% quota increase was predicted to reduce prices by 3.2%, while the actual price fell by 3.5%, showing excellent agreement in both direction and magnitude.

2. **Category B**: The model correctly predicted a price decline following a 5.9% quota increase, although the actual drop (1.1%) was smaller than the predicted 2.5% due to stronger demand, reflected by the higher bid-to-quota ratio (1.30 → 1.38).

3. **Overall**: The model reliably captures the direction and approximate magnitude of quota-driven price changes. Differences from actual outcomes are largely explained by concurrent changes in demand, making it best suited as a policy decision-support tool rather than an exact price forecasting model.